

Potential Talents - Technical Report

Candidate ranking, relevance validation, semantic retrieval, and recruiter feedback

0.948 NDCG@10 - Aspiring HR	0.935 NDCG@10 - Seeking HR	100% Token occurrence coverage	52 Unique titles evaluated
---------------------------------------	--------------------------------------	--	--------------------------------------

Executive Summary

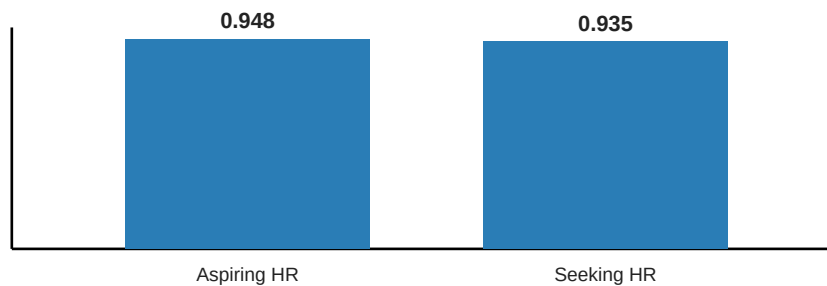
The workflow removes duplicate titles before evaluation, establishes an auditable relevance benchmark, ranks candidates semantically, and then adapts the ranking using recruiter feedback. The verified run achieved NDCG@10 of **0.948** and **0.935**, with complete vocabulary coverage.

Project Objectives

Overall objective: Develop a transparent and adaptable candidate-ranking system that helps recruiters identify the most relevant candidates faster and more consistently.

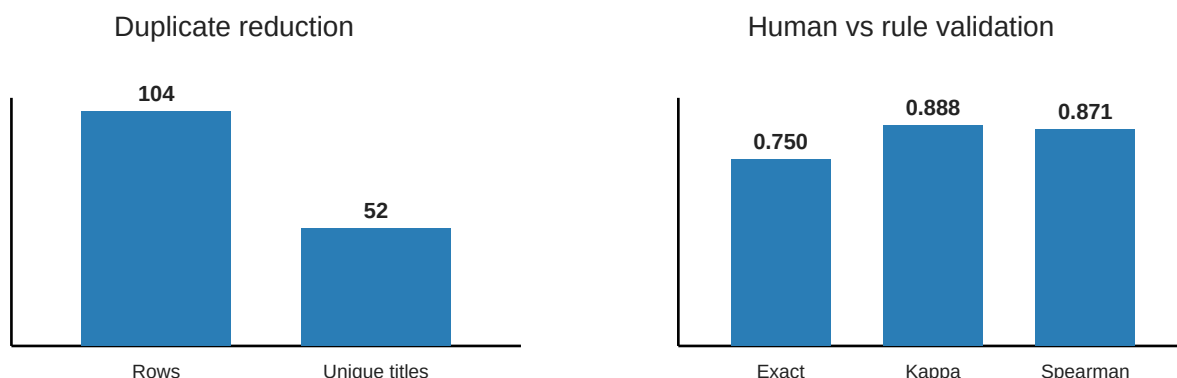
- 1. Remove duplicate job titles and analyse title vocabulary** to prevent repeated profiles from biasing results and support reliable candidate ranking.
- 2. Define explicit relevance rules for aspiring human resources and seeking human resources** to reduce false positives and make automated screening auditable.
- 3. Validate the relevance rules against independent human labels** to confirm alignment with recruiter judgement and reduce manual review effort.
- 4. Rank candidates by semantic similarity to recruiter search intent** to identify relevant candidates despite differences in job-title wording and improve candidate discovery.
- 5. Measure ranking quality using NDCG@10** to confirm the strongest candidates appear first and reduce recruiter review time.
- 6. Test recruiter-feedback strength against candidate movement and top-10 retention** to personalize recommendations at scale without unnecessarily disrupting strong existing candidates.

Stage-1 ranking quality



1. Data Audit and Relevance Reference

The source contains 104 candidate rows but only 52 unique job titles. Ranking each unique title once prevents repeated text from receiving disproportionate weight. Fourteen titles repeat and the largest exact-title frequency is seven.



Rule-based relevance

The query-specific reference is $R = H(0.70 + 0.30I)$. H is occupational relevance and I is intent alignment. Multiplying intent by occupational relevance prevents generic seeking-employment wording from becoming HR-relevant on intent alone. Clear employer solicitations receive zero candidate relevance.

Validation measure	Result
Exact ordinal agreement	39/52 (75.0%)
Quadratic weighted Cohen's kappa	0.888
Spearman correlation	0.871

After deduplication, 31/52 titles explicitly contain Human Resources, 34/52 contain direct HR evidence, 12 contain aspiring, and 10 contain seeking. This supports an auditable rule layer plus semantic ranking.

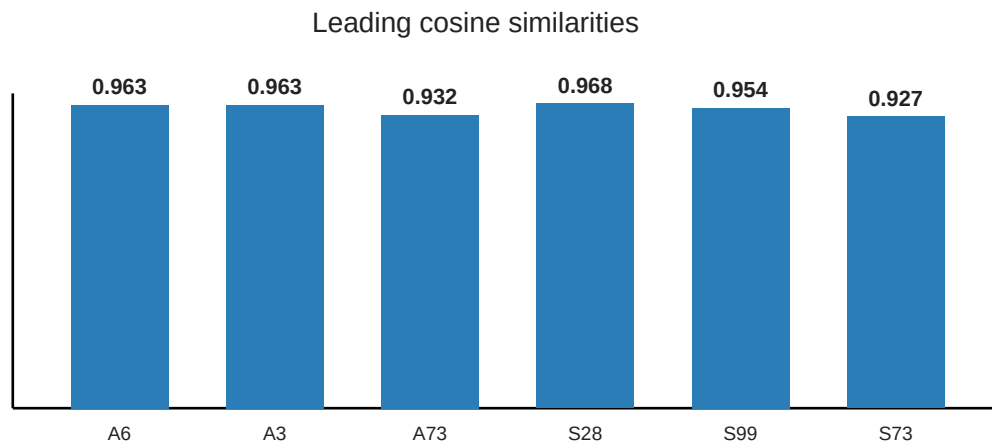
2. Semantic Ranking

The implementation uses **NLPL model 40 - English CoNLL17 Word2Vec 100D, continuous skip-gram**. Vocabulary header: 4,027,169; official archive approximately 3.03 GB. Candidate titles and recruiter queries are represented by mean word vectors and ranked using cosine similarity.

Coverage quantity	Result
Meaningful title-token occurrences	368
Title + query occurrences	374
Resolved token occurrences	374 (100%)
Unique meaningful tokens	208
Unique OOV tokens	0

Stage-1 results

NDCG@10 is **0.948** for aspiring HR and **0.935** for seeking HR, mean **0.942**. The leading aspiring IDs are 6, 3, 73; the leading seeking IDs are 28, 99, 73.

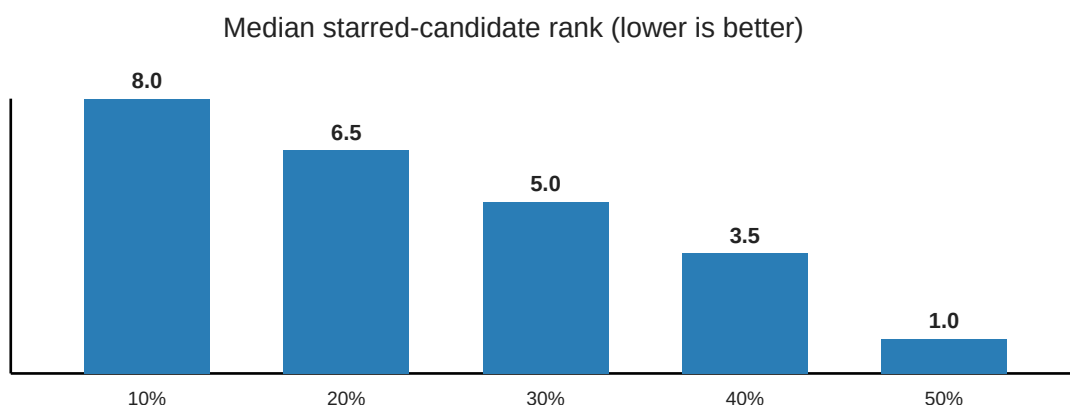


NDCG@10 values close to 1.0 indicate that the most relevant titles appear near the top of the semantic ranking. Complete token coverage also avoids missing candidate evidence because of vocabulary gaps in this dataset.

3. Recruiter Feedback and Reranking

A starred candidate shifts the query as $q_2 = \text{normalize}((1-w)q + w \cdot d^*)$. The experiment evaluates 34 strong title/query scenarios at five feedback weights.

Feedback	Median starred rank	Median top-10 retained
10%	8.0	9/10
20%	6.5	9/10
30%	5.0	9/10
40%	3.5	8/10
50%	1.0	8/10



Interpretation

No tested weight satisfies the former joint target of median starred rank ≤ 2 while retaining at least 9/10 of the original top ten. The result is therefore a personalization-stability trade-off, not a universal optimum. At 30%, both representative starred candidates move 2 -> 1 while retaining 9/10 and 10/10 of the original top ten.

Query / candidate	Rank	Top-10 retained	NDCG@10
Aspiring HR - ID 3	2 -> 1	9/10	0.948 -> 0.955
Seeking HR - ID 99	2 -> 1	10/10	0.935 -> 0.940

4. Business Significance, Limitations and Reproducibility

The business value comes from combining consistent screening with adaptable ranking. Deduplication avoids over-weighting repeated profiles; explicit relevance rules reduce false positives and make automated screening reviewable; semantic ranking broadens candidate discovery; NDCG@10 tests whether recruiter attention is focused near the best candidates; and feedback weighting allows personalization without retraining the model.

Limitation	Business implication
Mean pooling	Long titles can dilute important HR terms.
Limited context	Word2Vec cannot fully resolve directionality or sentence meaning.
Title-only feature	Skills and experience absent from the title cannot affect ranking.
Rule portability	New occupations require new role/intent definitions and validation.
Feedback strength	Higher weight moves starred candidates up faster but changes more of the shortlist.

Reproducibility

Notebook 01 runs from the included CSV files. Notebooks 02 and 03 require the official NLPL 40.zip archive and stream only the vectors needed for the 52 titles and two queries. Official archive: <https://vectors.nlpl.eu/repository/20/40.zip>